# MSG No. 136.506 $P_I4_2/mnm$ [ Type IV, tetragonal ]

Table 1: Wyckoff site: `2a`, site symmetry: `4'/mm'm`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, 0]$ | `[1,6,7,8,9,14,15,16,18,19,20,21,26,27,28,29]` |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[2,3,4,5,10,11,12,13,17,22,23,24,25,30,31,32]` |

Table 2: Wyckoff site: `2b`, site symmetry: `4'/mm'm`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, 0, \frac{1}{2}\right]$ | `[1,6,7,8,9,14,15,16,18,19,20,21,26,27,28,29]` |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | `[2,3,4,5,10,11,12,13,17,22,23,24,25,30,31,32]` |

Table 3: Wyckoff site: `4c`, site symmetry: `mm'm'.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, 0\right]$ | `[1,6,9,14,20,21,28,29]` |
| 2 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | `[2,3,10,11,23,24,31,32]` |
| 3 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | `[4,5,12,13,17,22,25,30]` |
| 4 | $\left[\frac{1}{2}, 0, 0\right]$ | `[7,8,15,16,18,19,26,27]` |

Table 4: Wyckoff site: `4d`, site symmetry: `-4m'2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, \frac{1}{4}\right]$ | `[1,6,10,11,23,24,28,29]` |
| 2 | $\left[0, \frac{1}{2}, \frac{3}{4}\right]$ | `[2,3,9,14,20,21,31,32]` |
| 3 | $\left[\frac{1}{2}, 0, \frac{1}{4}\right]$ | `[4,5,15,16,18,19,25,30]` |
| 4 | $\left[\frac{1}{2}, 0, \frac{3}{4}\right]$ | `[7,8,12,13,17,22,26,27]` |

Table 5: Wyckoff site: `4e`, site symmetry: `4'm'm`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, z]$ | `[1,6,15,16,18,19,28,29]` |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$ | `[2,3,12,13,17,22,31,32]` |
| 3 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$ | `[4,5,10,11,23,24,25,30]` |
| 4 | $[0, 0, -z]$ | `[7,8,9,14,20,21,26,27]` |

Table 6: Wyckoff site: `8f`, site symmetry: `..2'/m`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[1,16,24,25]` |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[2,12,20,26]` |
| 3 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[3,13,21,27]` |

*continued ...*

Table 6

| No. | position | mapping |
|---|---|---|
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | [4,10,18,28] |
| 5 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | [5,11,19,29] |
| 6 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | [6,15,23,30] |
| 7 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | [7,14,22,31] |
| 8 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | [8,9,17,32] |

Table 7: Wyckoff site: 8g, site symmetry: 2m'm'.

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, z\right]$ | [1,6,28,29] |
| 2 | $\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$ | [2,3,31,32] |
| 3 | $\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$ | [4,5,25,30] |
| 4 | $\left[\frac{1}{2}, 0, -z\right]$ | [7,8,26,27] |
| 5 | $\left[0, \frac{1}{2}, -z\right]$ | [9,14,20,21] |
| 6 | $\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$ | [10,11,23,24] |
| 7 | $\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$ | [12,13,17,22] |
| 8 | $\left[\frac{1}{2}, 0, z\right]$ | [15,16,18,19] |

Table 8: Wyckoff site: 8h, site symmetry: m.2m

| No. | position | mapping |
|---|---|---|
| 1 | $[x, x, 0]$ | [1,7,14,16] |
| 2 | $\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}\right]$ | [2,5,11,12] |
| 3 | $\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$ | [3,4,10,13] |
| 4 | $[-x, -x, 0]$ | [6,8,9,15] |
| 5 | $\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$ | [17,23,30,32] |
| 6 | $[-x, x, 0]$ | [18,21,27,28] |
| 7 | $[x, -x, 0]$ | [19,20,26,29] |
| 8 | $\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}\right]$ | [22,24,25,31] |

Table 9: Wyckoff site: 8i, site symmetry: m2'm'.

| No. | position | mapping |
|---|---|---|
| 1 | $[x, 0, 0]$ | [1,14,20,29] |
| 2 | $\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$ | [2,11,23,32] |
| 3 | $\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$ | [3,10,24,31] |
| 4 | $\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | [4,13,17,30] |
| 5 | $\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$ | [5,12,22,25] |
| 6 | $[-x, 0, 0]$ | [6,9,21,28] |
| 7 | $[0, x, 0]$ | [7,16,18,27] |
| 8 | $[0, -x, 0]$ | [8,15,19,26] |

Table 10: Wyckoff site: `8j`, site symmetry: `m2'm'`.

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{2},\ 0\right]$ | `[1,14,20,29]` |
| 2 | $\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | `[2,11,23,32]` |
| 3 | $\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | `[3,10,24,31]` |
| 4 | $\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$ | `[4,13,17,30]` |
| 5 | $\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$ | `[5,12,22,25]` |
| 6 | $\left[-x,\ \frac{1}{2},\ 0\right]$ | `[6,9,21,28]` |
| 7 | $\left[\frac{1}{2},\ x,\ 0\right]$ | `[7,16,18,27]` |
| 8 | $\left[\frac{1}{2},\ -x,\ 0\right]$ | `[8,15,19,26]` |

Table 11: Wyckoff site: `16k`, site symmetry: `..2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$ | `[1,23]` |
| 2 | $\left[-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$ | `[2,21]` |
| 3 | $\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$ | `[3,20]` |
| 4 | $\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$ | `[4,19]` |
| 5 | $\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$ | `[5,18]` |
| 6 | $\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$ | `[6,24]` |
| 7 | $\left[x+\frac{1}{2},\ x,\ \frac{3}{4}\right]$ | `[7,17]` |
| 8 | $\left[\frac{1}{2}-x,\ -x,\ \frac{3}{4}\right]$ | `[8,22]` |
| 9 | $\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$ | `[9,31]` |
| 10 | $\left[x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$ | `[10,29]` |
| 11 | $\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$ | `[11,28]` |
| 12 | $\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$ | `[12,27]` |
| 13 | $\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$ | `[13,26]` |
| 14 | $\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$ | `[14,32]` |
| 15 | $\left[\frac{1}{2}-x,\ -x,\ \frac{1}{4}\right]$ | `[15,25]` |
| 16 | $\left[x+\frac{1}{2},\ x,\ \frac{1}{4}\right]$ | `[16,30]` |

Table 12: Wyckoff site: `16l`, site symmetry: `m..`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ 0]$ | `[1,14]` |
| 2 | $\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | `[2,11]` |
| 3 | $\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | `[3,10]` |
| 4 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | `[4,13]` |
| 5 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$ | `[5,12]` |
| 6 | $[-x,\ -y,\ 0]$ | `[6,9]` |
| 7 | $[y,\ x,\ 0]$ | `[7,16]` |
| 8 | $[-y,\ -x,\ 0]$ | `[8,15]` |
| 9 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$ | `[17,30]` |
| 10 | $[-y,\ x,\ 0]$ | `[18,27]` |
| 11 | $[y,\ -x,\ 0]$ | `[19,26]` |
| 12 | $[x,\ -y,\ 0]$ | `[20,29]` |

*continued ...*

Table 12

| No. | position | mapping |
|---|---|---|
| 13 | $[-x,\ y,\ 0]$ | [21,28] |
| 14 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | [22,25] |
| 15 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | [23,32] |
| 16 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | [24,31] |

Table 13: Wyckoff site: 16m, site symmetry: ..m

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ x,\ z]$ | [1,16] |
| 2 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [2,12] |
| 3 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | [3,13] |
| 4 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | [4,10] |
| 5 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [5,11] |
| 6 | $[-x,\ -x,\ z]$ | [6,15] |
| 7 | $[x,\ x,\ -z]$ | [7,14] |
| 8 | $[-x,\ -x,\ -z]$ | [8,9] |
| 9 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [17,32] |
| 10 | $[-x,\ x,\ z]$ | [18,28] |
| 11 | $[x,\ -x,\ z]$ | [19,29] |
| 12 | $[x,\ -x,\ -z]$ | [20,26] |
| 13 | $[-x,\ x,\ -z]$ | [21,27] |
| 14 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | [22,31] |
| 15 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [23,30] |
| 16 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | [24,25] |

Table 14: Wyckoff site: 16n, site symmetry: .m'.

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ y,\ z]$ | [1,28] |
| 2 | $\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | [2,31] |
| 3 | $\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | [3,32] |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$ | [4,25] |
| 5 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [5,30] |
| 6 | $[0,\ -y,\ z]$ | [6,29] |
| 7 | $[y,\ 0,\ -z]$ | [7,26] |
| 8 | $[-y,\ 0,\ -z]$ | [8,27] |
| 9 | $[0,\ -y,\ -z]$ | [9,20] |
| 10 | $\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | [10,23] |
| 11 | $\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | [11,24] |
| 12 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [12,17] |
| 13 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | [13,22] |
| 14 | $[0,\ y,\ -z]$ | [14,21] |
| 15 | $[-y,\ 0,\ z]$ | [15,18] |
| 16 | $[y,\ 0,\ z]$ | [16,19] |

Table 15: Wyckoff site: 32o, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | [1] |
| 2 | $\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [2] |
| 3 | $\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | [3] |
| 4 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$ | [4] |
| 5 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [5] |
| 6 | $[-x,\ -y,\ z]$ | [6] |
| 7 | $[y,\ x,\ -z]$ | [7] |
| 8 | $[-y,\ -x,\ -z]$ | [8] |
| 9 | $[-x,\ -y,\ -z]$ | [9] |
| 10 | $\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | [10] |
| 11 | $\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [11] |
| 12 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [12] |
| 13 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | [13] |
| 14 | $[x,\ y,\ -z]$ | [14] |
| 15 | $[-y,\ -x,\ z]$ | [15] |
| 16 | $[y,\ x,\ z]$ | [16] |
| 17 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [17] |
| 18 | $[-y,\ x,\ z]$ | [18] |
| 19 | $[y,\ -x,\ z]$ | [19] |
| 20 | $[x,\ -y,\ -z]$ | [20] |
| 21 | $[-x,\ y,\ -z]$ | [21] |
| 22 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | [22] |
| 23 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [23] |
| 24 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | [24] |
| 25 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$ | [25] |
| 26 | $[y,\ -x,\ -z]$ | [26] |
| 27 | $[-y,\ x,\ -z]$ | [27] |
| 28 | $[-x,\ y,\ z]$ | [28] |
| 29 | $[x,\ -y,\ z]$ | [29] |
| 30 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [30] |
| 31 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | [31] |
| 32 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | [32] |